Forecast final performance from available evidence

Completed: reproducible retrospective study | Next: timestamped, sequential forecasting

191
matched student-course rows

6
terms: Fall 2024–Spring 2026

22
inputs in the selected model

Which variables improve final-grade forecasts when they are observable by a defined cutoff?

1 AUDIT
Normalize, deduplicate, join

2 MODEL
Fixed DAG + Bayesian CPDs

3 VALIDATE
Earlier → later course terms

Scope: association-based prediction; causal effects and coaching effectiveness are not estimated.

One student-term row; explicit feature eligibility

Completed: source-column manifest, points normalization, LMS deduplication and train-only preprocessing

FEATURE CONSTRUCTION

Block A: 16 raw marks → 10 grade features
Block B: 23 raw marks → 16 grade features

Both add 4 survey constructs + 2 log-LMS signals.

EARLY SURVEY + LMS

First four assumed course weeks
Survey missing: 78 / 191
LMS missing: 2 / 191

Released-assessment timing is not verified.

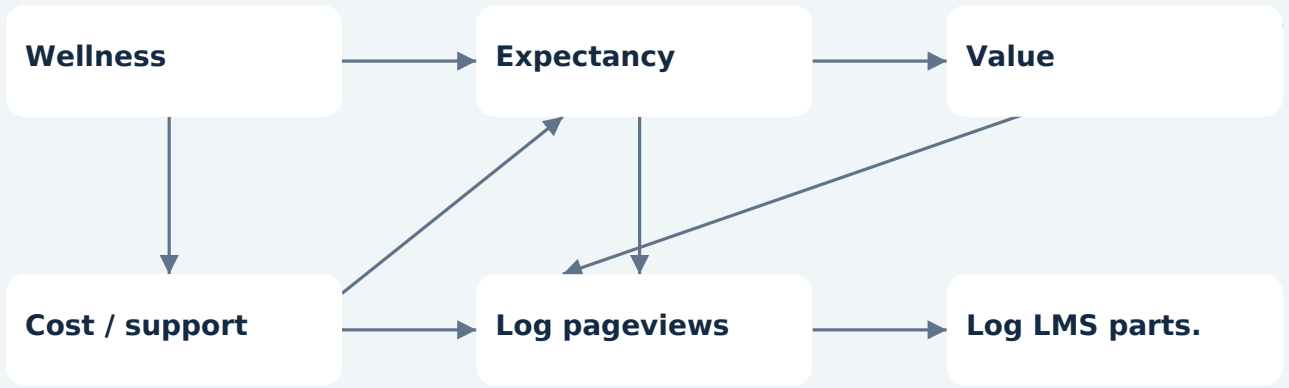
Final/current totals, final quiz and Group B outcomes are excluded.
Final exports can still contain later revisions of earlier marks.

Interpretation: retrospective evidence bundles; actual prediction dates and intervention lead time remain unknown.

A reviewable 23-node, 62-edge Gaussian DAG

Arrows encode assumed conditional relationships; the data estimate their strengths.

PREDICTOR SUBGRAPH / 8 EDGES



GRADE LAYER / 32 EDGES

Expectancy + log pageviews
→ each of 16 grade nodes

OUTCOME LAYER / 22 EDGES

6 survey/LMS + 16 grades
→ final course percentage

Joint factorization: $p(x_1, \ldots, x_{23}) = \prod_j p(x_j \mid \text{parents}_j)$

Current graph does not explicitly model MLC exposure/use, latent ability, section effects or response selection.

Learn local CPDs; condition on available evidence

Completed: Normal–Inverse-Gamma shrinkage | Added: exact outcome Student-t check for complete evidence

$$z_j \mid z_{pa(j)}, \beta_j, \sigma_j^2 \sim \mathcal{N}(a_j + z_{pa(j)}^T \beta_j, \sigma_j^2)$$
$$\beta_j \mid \sigma_j^2 \sim \mathcal{N}(0, \sigma_j^2 \Lambda_0^{-1}), \quad \sigma_j^2 \sim IG(2, 1)$$

01

FIT

Train-only median filling / scaling
Coefficient precisions: 1 or 10
Posterior mean CPD parameters

02

INFER

Condition joint Gaussian on inputs
Marginalize missing evidence
Return final-% mean and interval

03

AUDIT

Plug-in ≠ full posterior prediction
New Student-t CPD diagnostic
Complete predictor evidence only

New evidence updates a student forecast; new labeled outcomes update parameters. These are different operations.

Useful retrospective forecasts; no added signal shown

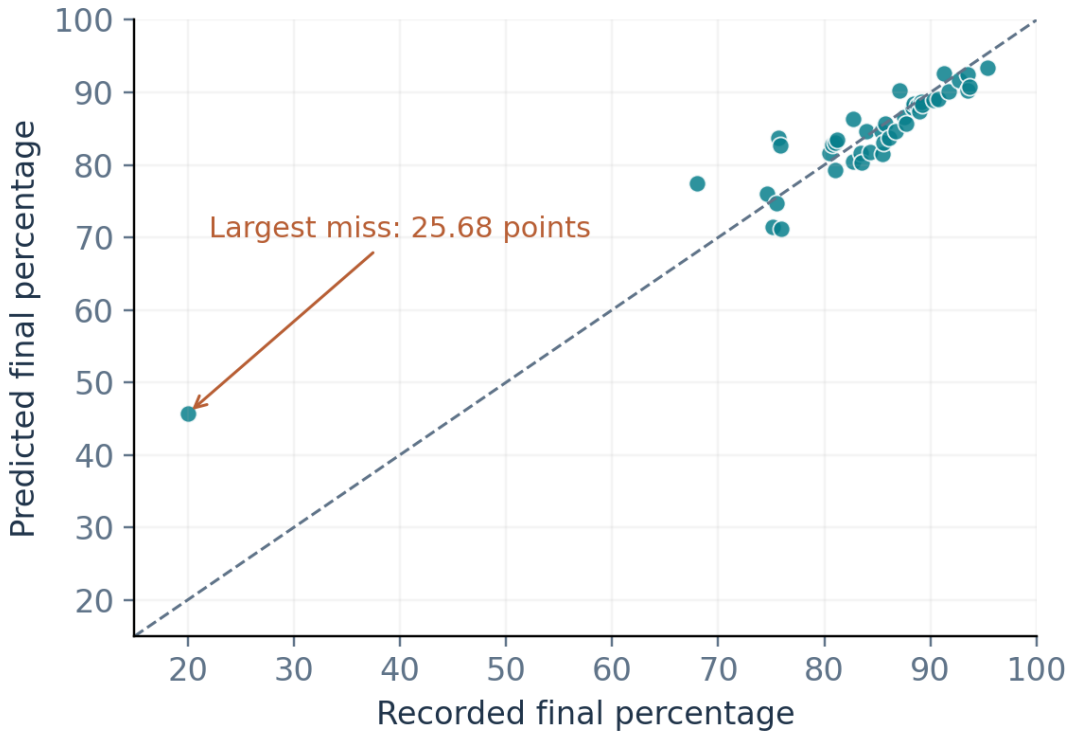
Selected on 3 earlier-term folds: Block B + precision 10; macro validation MAE = 3.09 points.

148 earlier students → 43 Spring 2026 students

Spring 2026 is a reused diagnostic cohort.

MODEL / COMMON EVALUATION COHORT	MAE
Full DAG model	**2.83**
Grades-only benchmark	2.82
Train-mean baseline	8.13

RMSE 4.92 • R^2 0.817
90% interval: 39/43 covered; width 10.02 points



All errors are percentage points. Small cohorts, final-grade component overlap and unverified timing limit claims.

Do wellness, motivation and engagement add value?

Hypothesis: these signals may help earlier, before assessment history dominates; this remains untested.

TIME-MATCHED FEATURE COMPARISONS

Baseline	Released grades only
+ Wellness	Same grades + wellness
+ Motivation	Same grades + expectancy / value / cost
+ Engagement	Same grades + LMS signals
Full model	Same grades + all three groups

Same cutoff • same students • matched tuning
Add leave-one-group-out sensitivity checks.

EVIDENCE REQUIRED

1 Earlier → later terms

Train-only preprocessing; tune before evaluation.

2 Paired improvement + uncertainty

Report MAE reduction by term; CRPS + coverage / width.

3 Missingness + new-cohort check

Audit 78/191 missing surveys; freeze, then validate.

Success: a pre-defined useful gain, consistent across terms and confirmed on new students. A null result is valid.

Move from retrospective fit to a defensible early forecast

Proposed work below; the current notebook has not yet validated a prospective coaching workflow.

1

FREEZE THE PREDICTION DATE

Verify releases + score revisions; build immutable weekly snapshots.

2

REVIEW THE THEORY-INFORMED DAG

Add prior preparation, MLC exposure/use, context and response selection.

3

STRENGTHEN UNCERTAINTY + UPDATING

Missing-data model; term/section pooling; frozen-scale posterior updates.

4

VALIDATE ON A NEW COHORT

Time-matched ablations; MAE/RMSE, log score, coverage and interval width.

Discussion: which cutoff offers useful lead time, and what would count as convincing prospective evidence?